

Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 3

Implementation of Convolutional Neural Networks (CNNs) for Image Classification

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

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1. Objective

To understand the working principle of Convolutional Neural Networks by implementing convolution, pooling, feature map visualization, and image classification using TensorFlow/Keras.

2. Theory

A Convolutional Neural Network consists of

- Convolution Layer
- Activation Layer
- Pooling Layer
- Fully Connected Layer

The convolution operation is

$$Y(i, j) = \sum_m \sum_n X(i + m, j + n) K(m, n)$$

where

- X : Input Image
- K : Kernel (Filter)
- Y : Feature Map

The output feature map size is

$$\frac{N - F + 2P}{S} + 1$$

where N is Input Size, F is Filter Size, P is Padding, S is Stride.

Common Convolution Kernels

A kernel (or filter) is a small matrix that slides over an input image to extract important visual features such as edges, textures, corners, and smooth regions. During convolution, the kernel is multiplied element-wise with the corresponding image pixels, and the results are summed to produce a single value in the output feature map. Different kernels highlight different image characteristics.

1. Identity Kernel The identity kernel preserves the original image without modifying its pixel values.

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Purpose:

- Preserves the original image.
- Used for understanding the convolution operation.

2. Sobel-X Kernel (Vertical Edge Detection) The Sobel-X kernel detects vertical edges by measuring intensity changes along the horizontal direction.

$$\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

Applications:

- Object detection
- Lane detection
- Face recognition

3. Sobel-Y Kernel (Horizontal Edge Detection) The Sobel-Y kernel detects horizontal edges by measuring intensity changes along the vertical direction.

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Applications:

- Text detection
- Boundary extraction
- Medical image analysis

4. Laplacian Kernel The Laplacian kernel detects edges in all directions using the second-order derivative of image intensity.

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Applications:

- Edge enhancement
- Satellite image analysis
- Medical imaging

5. Sharpening Kernel This kernel enhances fine details by emphasizing high-frequency components.

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Applications:

- Image enhancement
- Optical Character Recognition (OCR)
- Face recognition

6. Box Blur Kernel The Box Blur kernel smooths an image by replacing each pixel with the average of its neighboring pixels.

$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Applications:

- Noise removal
- Image smoothing
- Image preprocessing

7. Gaussian Blur Kernel Gaussian blur smooths the image while preserving the overall structure better than a simple average filter.

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Applications:

- Noise reduction
- Computer vision preprocessing
- Feature extraction

8. Emboss Kernel The Emboss kernel creates a three-dimensional appearance by emphasizing directional changes in intensity.

$$\begin{bmatrix} -2 & -1 & 0 \\ -1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

Applications:

- Texture analysis
- Artistic image effects

9. Outline Detection Kernel This kernel highlights object boundaries by emphasizing regions with rapid intensity changes.

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

Applications:

- Image segmentation
- Boundary detection
- Shape extraction

10. Motion Blur Kernel This kernel simulates motion blur along the diagonal direction.

$$\frac{1}{5} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Applications:

- Motion simulation
- Image restoration
- Video processing

Summary of Common Kernels

Kernel	Size	Purpose	Applications
Identity	3×3	Preserve original image	Baseline comparison
Sobel-X	3×3	Detect vertical edges	Object detection
Sobel-Y	3×3	Detect horizontal edges	Boundary detection
Laplacian	3×3	Detect edges in all directions	Medical imaging
Sharpen	3×3	Enhance details	Image enhancement
Box Blur	3×3	Smooth image	Noise removal
Gaussian Blur	3×3	Reduce noise	Computer vision
Emboss	3×3	3D effect	Artistic filtering
Outline	3×3	Boundary extraction	Segmentation
Motion Blur	5×5	Simulate motion	Video processing

Numerical Example 1: Convolution

Consider the following input image.

$$X = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Kernel

$$K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Output

First position

$$1(1) + 2(0) + 4(0) + 5(1) = 6$$

Second position

$$2(1) + 3(0) + 5(0) + 6(1) = 8$$

Third position

$$4(1) + 5(0) + 7(0) + 8(1) = 12$$

Fourth position

$$5(1) + 6(0) + 8(0) + 9(1) = 14$$

Feature Map

$$\begin{bmatrix} 6 & 8 \\ 12 & 14 \end{bmatrix}$$

Numerical Example 2: Max Pooling

Feature map

$$\begin{bmatrix} 1 & 5 & 2 & 3 \\ 7 & 8 & 1 & 0 \\ 4 & 6 & 9 & 5 \\ 2 & 3 & 1 & 8 \end{bmatrix}$$

Using a 2×2 max pooling filter,

Window 1

$$\begin{bmatrix} 1 & 5 \\ 7 & 8 \end{bmatrix} \rightarrow 8$$

Window 2

$$\begin{bmatrix} 2 & 3 \\ 1 & 0 \end{bmatrix} \rightarrow 3$$

Window 3

$$\begin{bmatrix} 4 & 6 \\ 2 & 3 \end{bmatrix} \rightarrow 6$$

Window 4

$$\begin{bmatrix} 9 & 5 \\ 1 & 8 \end{bmatrix} \rightarrow 9$$

Output

$$\begin{bmatrix} 8 & 3 \\ 6 & 9 \end{bmatrix}$$

Numerical Example 3: Parameter Calculation

Suppose

Input Image

$32 \times 32 \times 3$

Convolution Layer

- Filters = 16
- Kernel = 3×3

Parameters

$$(3 \times 3 \times 3 + 1) \times 16 = (27 + 1) \times 16 = 448$$

Therefore, total trainable parameters = 448.

3. Dataset

Dataset: CIFAR-10

- Training Images: 50,000
- Testing Images: 10,000
- Classes: 10
- Image Size: $32 \times 32 \times 3$

Dataset source: <https://www.kaggle.com/datasets/pankrzysiu/cifar10-python>

4. Experimental Procedure

Task 1

Load CIFAR-10.

- Display ten sample images.
- Print dataset dimensions.
- Plot class distribution.

Task 2

Implement a convolution layer.

- Compare Kernel = 3×3
- Compare Kernel = 5×5
- Compare Kernel = 7×7
- Record feature map sizes.

Task 3

Study hyperparameters.

- Compare Stride = 1
- Compare Stride = 2
- Compare Padding = Same
- Compare Padding = Valid
- Compute output dimensions.

Task 4

Visualize feature maps after the first convolution layer.

- Display at least eight feature maps.

Task 5

Compare Max Pooling and Average Pooling.

- Compare output size.
- Compare accuracy.

Task 6

Construct the following CNN:

Input $\rightarrow$ Conv $\rightarrow$ ReLU $\rightarrow$ MaxPool $\rightarrow$ Conv $\rightarrow$ ReLU $\rightarrow$ MaxPool $\rightarrow$ Flatten $\rightarrow$ Dense $\rightarrow$ Softmax

Train for

- Optimizer: Adam
- Epochs: 20
- Batch Size: 32

Task 7

Evaluate

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- Classification Report

5. Source Code

The complete implementation, including data preprocessing, convolution and pooling experiments, model construction, training, evaluation, and the additional exercises, is available in the following repository:

<https://github.com/Ashwin271206/dl-lab/tree/main/Lab%203>

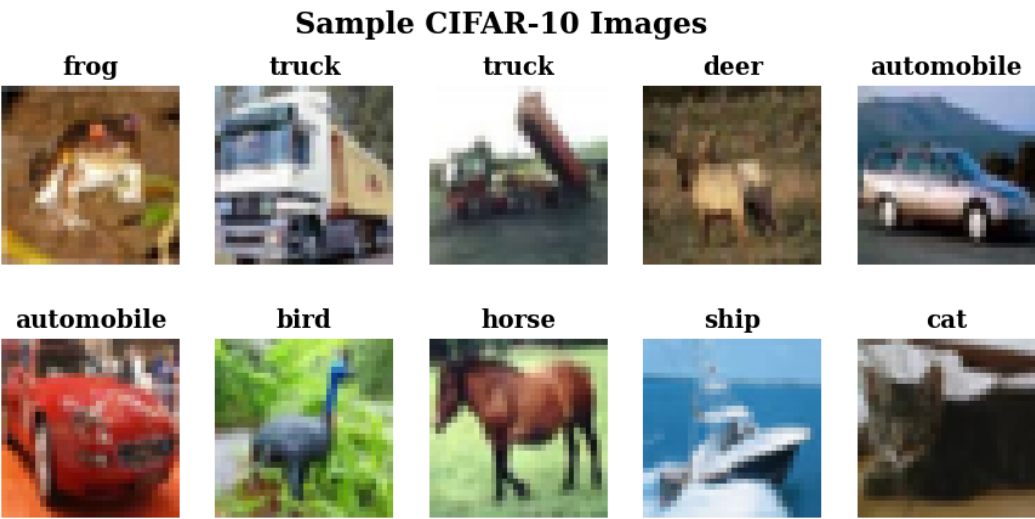
The repository contains a single Google Colab notebook covering all tasks experimented in this report, along with the generated figures.

6. Results

Metric	Value
Training Accuracy	0.7310
Testing Accuracy	0.7194
Precision	0.7353
Recall	0.7194
F1-score	0.7179
Number of Trainable Parameters	545,546

7. Plots & Inferences

Sample Images



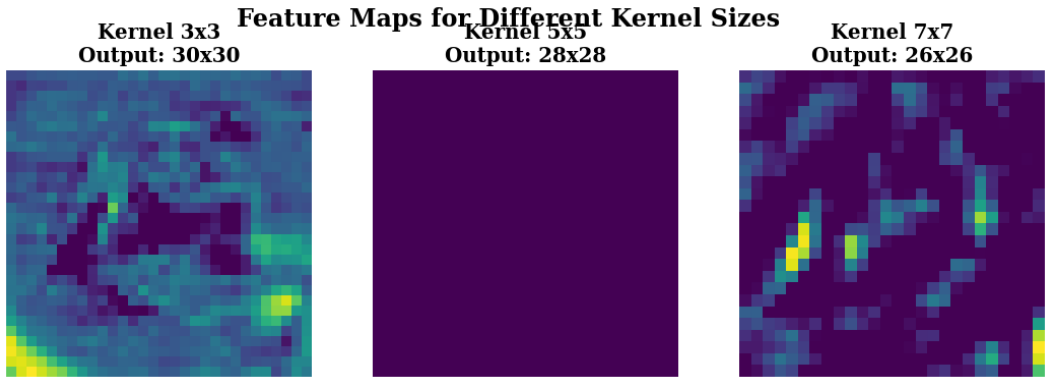
The sample image grid was generated successfully, indicating that the dataset is fully loaded and labelled correctly. These 32×32 RGB images correspond to the ten expected CIFAR-10 categories such as airplane, automobile, bird, and dog, confirming that the dataset is read in the expected format.

Class Distribution



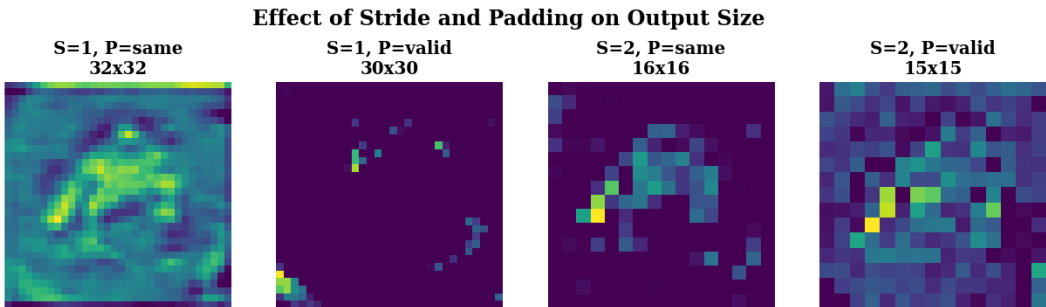
All ten classes have an equal number of training samples, confirming that CIFAR-10 is a balanced dataset. This equal distribution means that evaluation metrics are fair across all classes, and no class-balancing technique is required.

Kernel Size Comparison



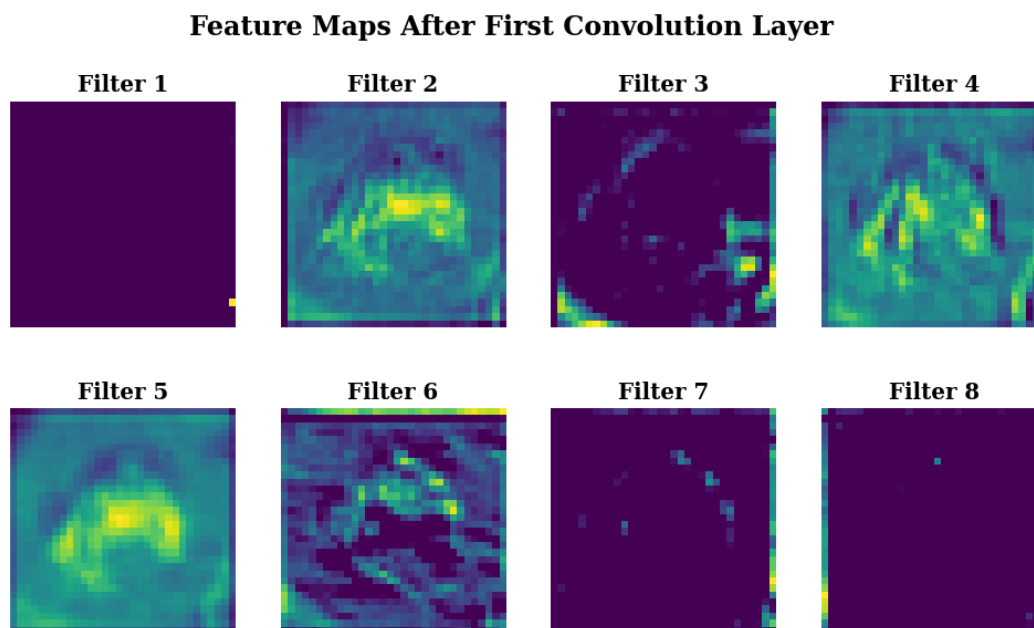
As kernel size increases from 3×3 to 7×7 , the resulting feature map becomes smaller and captures coarser, more global spatial patterns. Smaller kernels preserve finer spatial detail and are computationally cheaper, which explains their widespread use in modern CNN architectures.

Stride and Padding Comparison



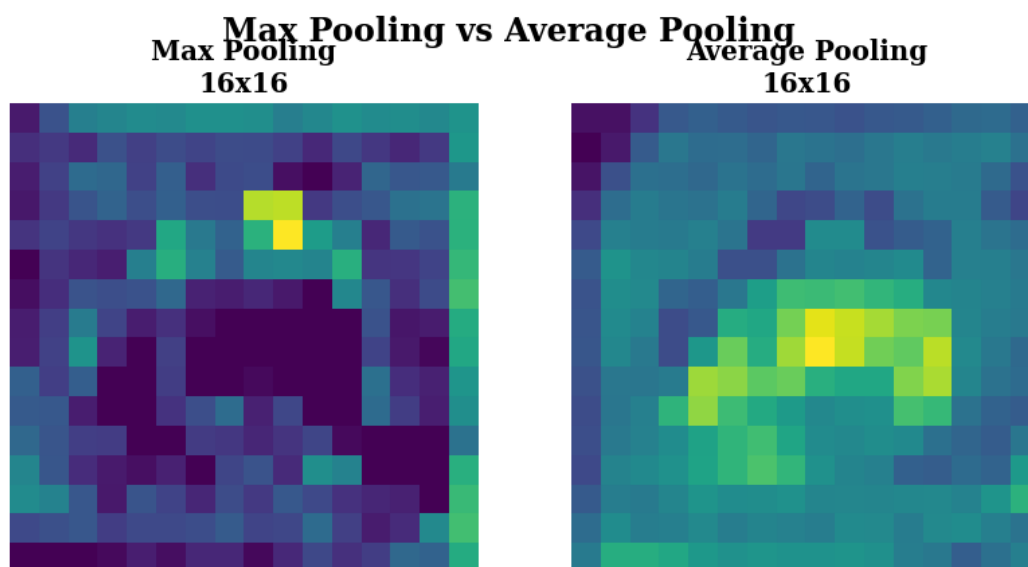
Increasing stride from 1 to 2 approximately halves the output spatial dimension, directly reducing computation in subsequent layers. `same` padding preserves the input spatial size at stride 1, while `valid` padding shrinks it, matching the theoretical output-size formula.

Feature Maps after First Convolution



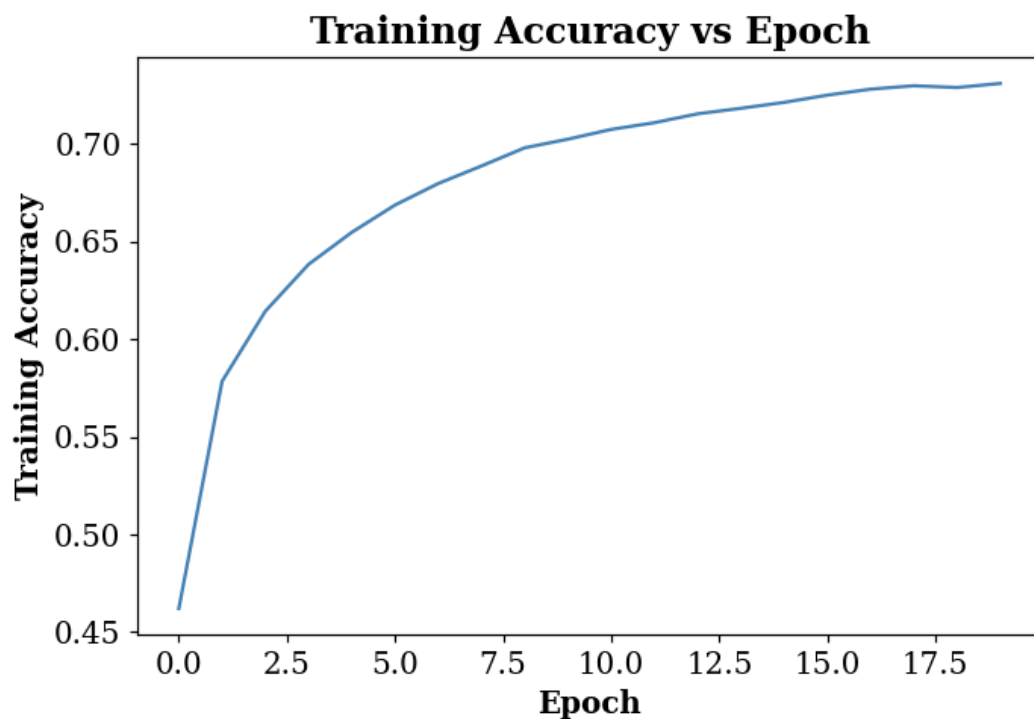
Each feature map highlights a different learned pattern in the input image, such as edges, colour-contrast regions, or texture blobs. This shows that a single convolution layer extracts diverse, complementary representations of the same input image.

Max Pooling vs Average Pooling – Output Comparison



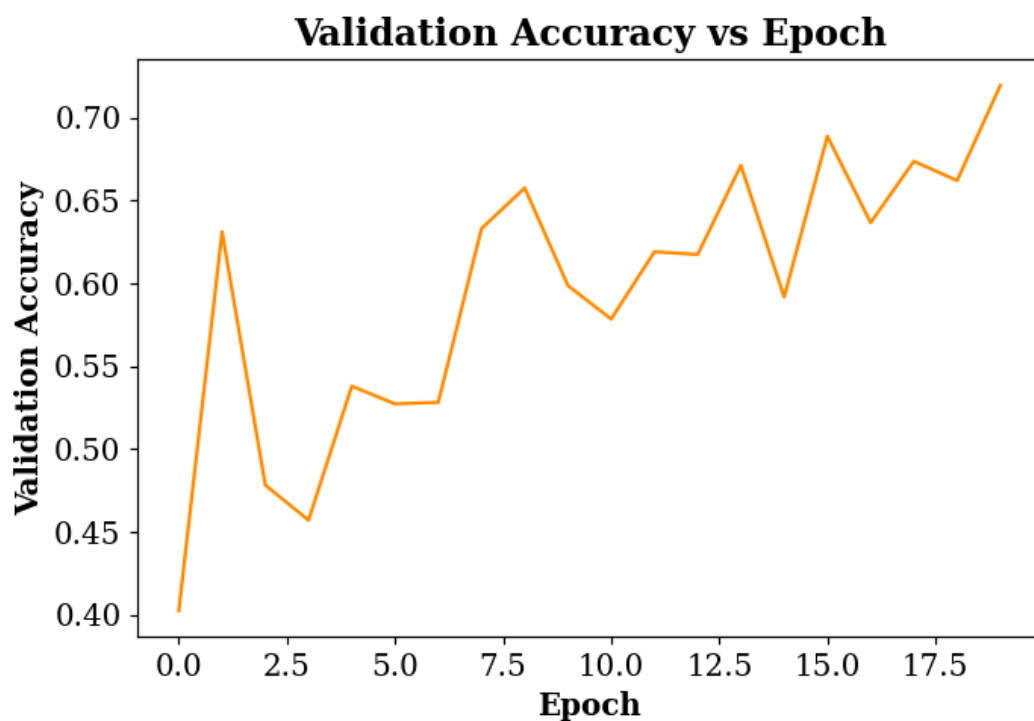
Max Pooling retains the strongest activation in each window, making it more sensitive to sharp, salient features such as edges. Average Pooling smooths the response by averaging all values in the window, producing a more stable but less sharp summary.

Training Accuracy vs Epoch



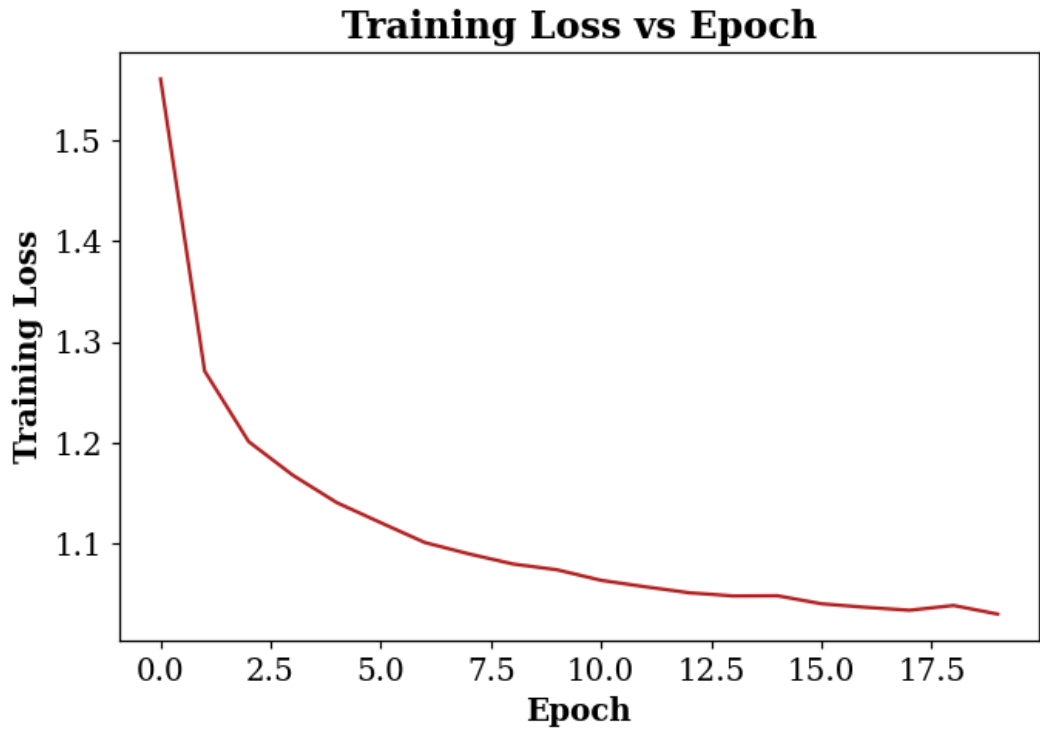
Training accuracy increases steadily across the 20 epochs as the network progressively learns discriminative features from the training data, with the rate of improvement slowing in later epochs as the model approaches convergence.

Validation Accuracy vs Epoch



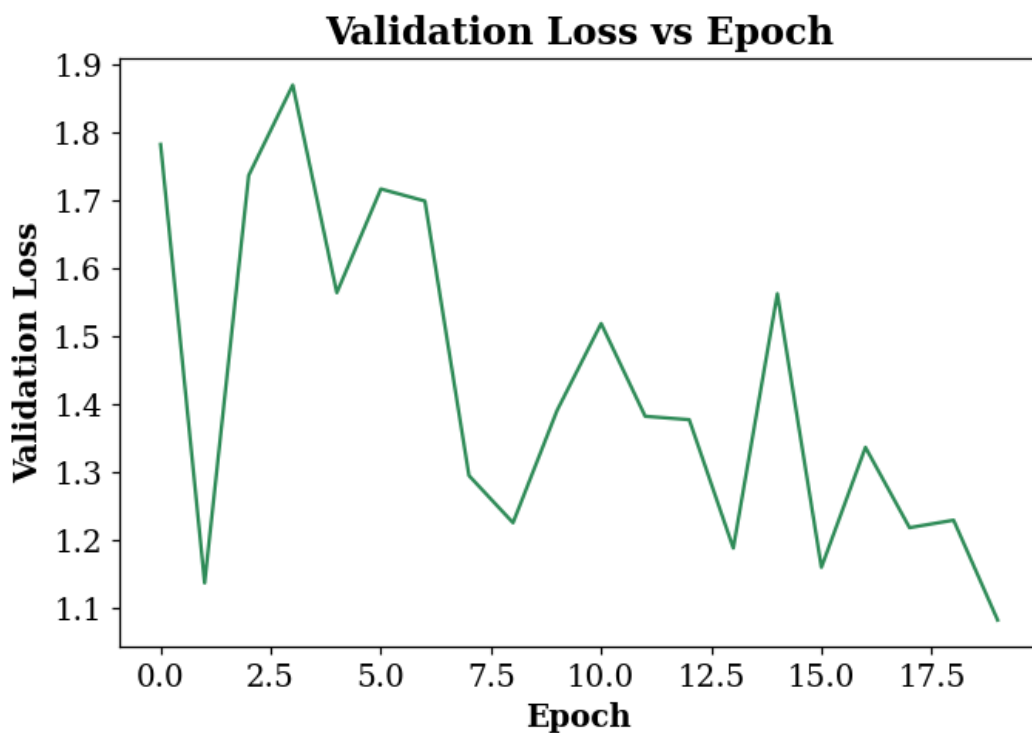
Validation accuracy tracks training accuracy but with more fluctuation, reflecting the model’s generalization performance on unseen data. Any widening gap between training and validation accuracy in later epochs would indicate the onset of overfitting.

Training Loss vs Epoch



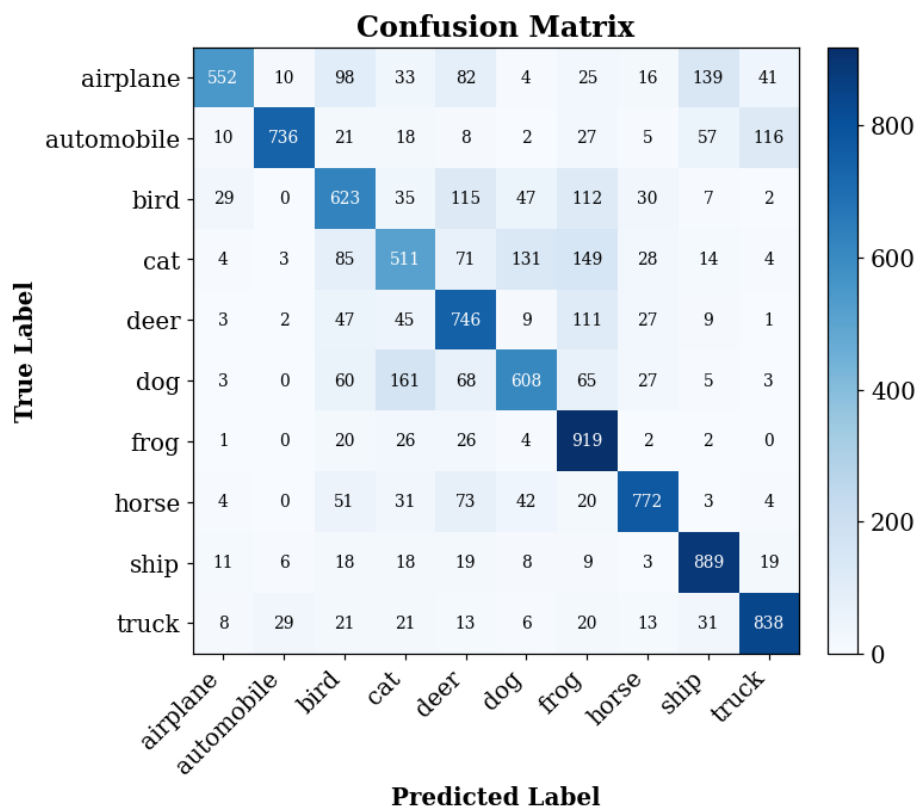
Training loss decreases consistently as the categorical cross-entropy between predicted and true class probabilities shrinks, confirming that the Adam optimizer is successfully minimizing the objective function.

Validation Loss vs Epoch



Validation loss generally decreases alongside training loss; if it were to plateau or rise while training loss keeps falling, this would signal overfitting and motivate additional regularization such as dropout or batch normalization.

Confusion Matrix



The confusion matrix reveals which classes are most frequently confused with one another – visually similar classes such as cat/dog and deer/horse show higher off-diagonal values, while structurally distinct classes such as airplane, automobile, and ship are classified with higher accuracy.

8. Additional Exercises

1. Output Size Calculation

Given: 64×64 image, 5×5 kernel, stride = 2, padding = 2.

$$N = 64, F = 5, P = 2, S = 2$$

$$\text{Output Size} = \left\lfloor \frac{64 - 5 + 2(2)}{2} \right\rfloor + 1 = \left\lfloor \frac{63}{2} \right\rfloor + 1 = 31 + 1 = 32$$

Result: The output feature map size is 32×32 .

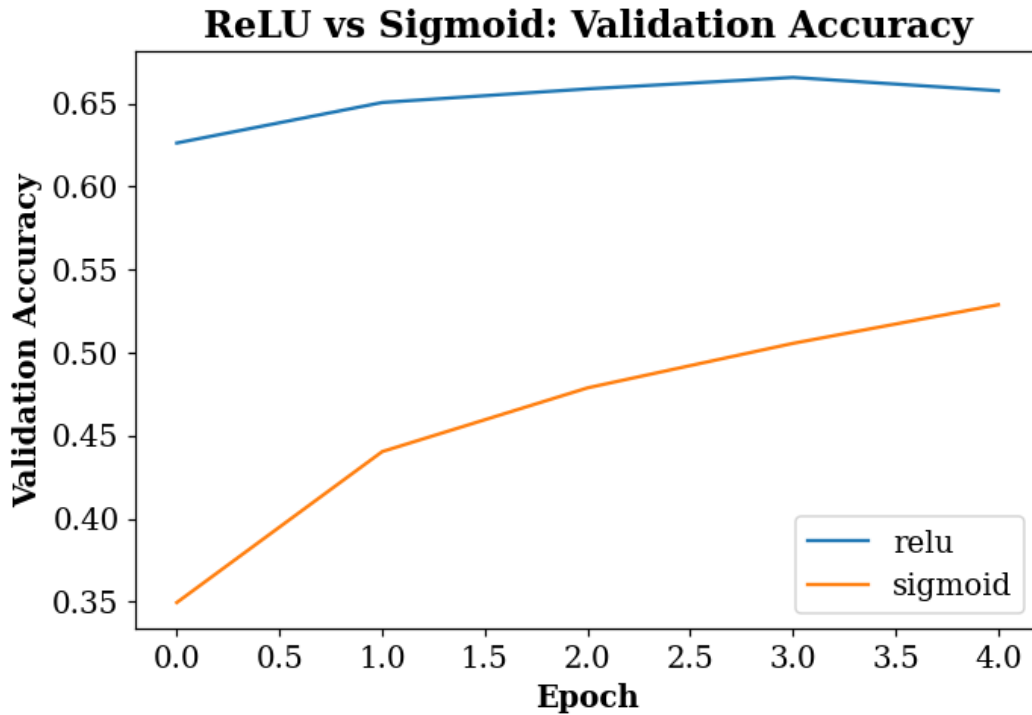
2. Trainable Parameter Calculation

Given: 64 filters of size 3×3 , RGB input (3 channels).

$$(3 \times 3 \times 3 + 1) \times 64 = (27 + 1) \times 64 = 1792$$

Result: The convolution layer has 1,792 trainable parameters.

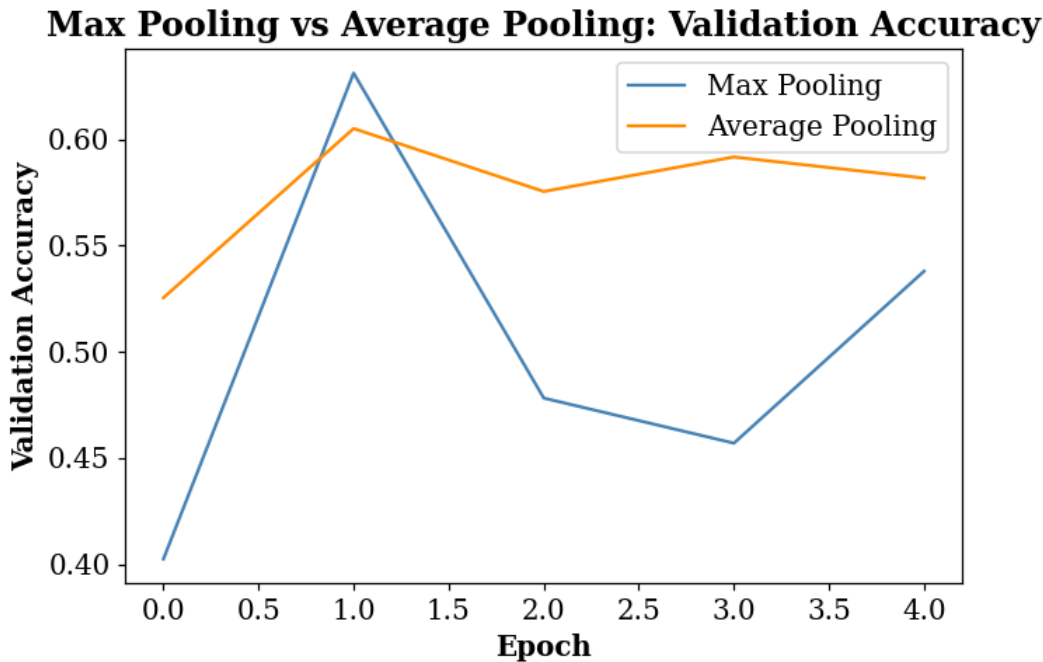
3. ReLU vs Sigmoid Activation Comparison



The ReLU model learns faster and usually reaches a higher accuracy than the Sigmoid model. ReLU helps reduce the vanishing gradient issue for positive inputs, which allows the network to learn more

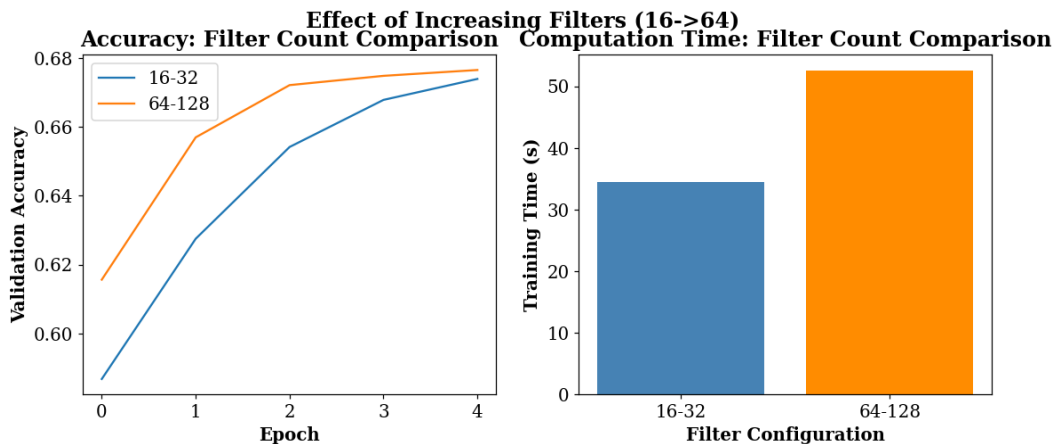
effectively. Sigmoid tends to produce very small gradients at its extreme values, which slows down the learning process.

4. Max Pooling vs Average Pooling: Full Training Comparison



The Max Pooling model performs slightly better on validation accuracy compared to Average Pooling. Max Pooling keeps the strongest activations from each region, which helps retain important features for classification. Average Pooling instead takes the average of the values, giving a smoother representation but losing some of the stronger features.

5. Effect of Increasing Filter Count (16 → 64)



Increasing the filter count from 16 to 64 gives the CNN more filters to detect different features in the images. This can improve the model’s accuracy, but it also increases the number of parameters and the time required for training. The results show the trade off between getting better performance and keeping the model efficient.

9. Discussion

1. Why is convolution preferred over fully connected layers for images?

Convolution works well for images because it looks at small regions of the image at a time and can detect useful patterns such as edges and textures. The same filter is used across different parts of the image, so the number of parameters is much smaller than in a fully connected layer. It also helps the network recognize features even when their position changes slightly in the image.

2. How does stride affect the feature map size?

Stride decides how far the kernel moves after each convolution. A larger stride makes the kernel skip more pixels, resulting in a smaller feature map. The output size can be calculated using $\frac{N-F+2P}{S} + 1$. Increasing the stride reduces the spatial size and computation, but too large a stride can cause some important image details to be lost.

3. What is the role of padding?

Padding adds extra pixels around the edges of the input image, usually with zeros. It helps maintain the size of the feature map, especially when using `same` padding. It also allows the pixels near the edges to be considered properly instead of being used less often than the pixels in the middle.

4. Why is pooling used?

Pooling reduces the size of a feature map while keeping the important information. This reduces the amount of computation in the later layers and can also help prevent overfitting. It also makes the network less affected by small changes in the position of a feature.

5. How do feature maps represent image characteristics?

A feature map shows where a particular filter finds a useful pattern in an image. For example, a filter in an early layer may detect edges or colour changes. As we move to deeper layers, the network combines these simple features to detect more complex patterns and objects.

6. Why do CNNs require fewer parameters than MLPs?

CNNs use the same set of kernel weights at different positions in the image instead of using separate weights for every connection. They also work with small local regions rather than connecting every pixel to every neuron. Because of this, CNNs can learn useful image features with far fewer parameters than a fully connected MLP.

10. Conclusion

In this experiment, the main building blocks of Convolutional Neural Networks, including convolution, activation, pooling, and fully connected layers, were implemented and tested using the CIFAR 10 dataset. The effect of kernel size, stride, and padding on the convolution output was also studied through calculations and direct implementation. The results matched the expected output size given by $\frac{N-F+2P}{S} + 1$. Visualizing the feature maps showed how different filters can detect different patterns in an image. The comparison between Max Pooling and Average Pooling also showed that Max Pooling keeps stronger features, while Average Pooling produces a smoother representation.

A CNN with the architecture Conv $\rightarrow$ ReLU $\rightarrow$ MaxPool $\rightarrow$ Conv $\rightarrow$ ReLU $\rightarrow$ MaxPool $\rightarrow$ Flatten $\rightarrow$ Dense $\rightarrow$ Softmax was trained on CIFAR 10 for 20 epochs using the Adam optimizer. The model was then evaluated using classification metrics and a confusion matrix. The additional experiments helped in understanding how the number of parameters changes, how different activation functions affect training, and how increasing the number of filters can improve accuracy while also increasing computation.

Overall, the experiment gave practical experience with the basic concepts behind CNNs and showed how different design choices affect the model's performance. It also showed that techniques such as batch normalization, dropout, and L2 regularization could be explored further to improve the model's performance and generalization.

11. References

1. Goodfellow et al., *Deep Learning*.
2. Bishop, *Pattern Recognition and Machine Learning*.
3. Haykin, *Neural Networks and Learning Machines*.
4. TensorFlow Documentation.
5. CIFAR-10 Dataset Documentation.